

Deep Learning and ANI-1 Accuracy

Introduction

Accurate and efficient prediction of molecular properties and energies is a fundamental challenge and point of interest in computational chemistry and materials sciences. There has been an increasing gap between information stores and the knowledge captured and utility achieved¹. Therefore, more efficient approaches to accelerate this conversion need to be explored. Machine learning (ML) has provided exceptional applications to uncover complexities and extract insights from data for density functional theory (DFT). Specifically, deep learning and neural networks have been a particularly powerful method for quantum mechanical (QM) DFT calculations for organic molecules².

A prominent example is the ANI-1 dataset, a comprehensive collection of molecular structures and energies generated using high-quality DFT calculations, in which artificial neural network (ANN) models can be trained on to accurately reproduce molecular energies and forces, approaching the accuracy of calculations². Smith et al. presented strong neural network potential (NNP) for organic molecules using machine learning and deep learning architecture¹. They expanded upon Behler and Perrinello's high-dimensional neural network (HDNN) method to build potentials for organic molecules with further application to other classes of molecules or even crystals³. Their study revealed that the ANI method is not only effective and accurate in molecular simulation, but performs exceptionally faster computationally than through DFT (Figure 1)¹.

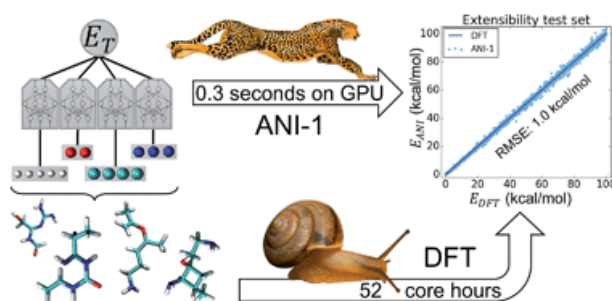


Figure 1. ANI-1 versus DFT computational efficiency.

This project attempts to use knowledge from previous work and improve upon these findings to enhance model performance. I studied how hyperparameter tuning and architectural changes can improve the performance of an ANN on the ANI-1 dataset, and compared these insights with the original paper.

Methods

Data preparation was conducted using TorchANI, each neural network was made using PyTorch, N-fold cross-validation was performed with Sklearn, and visualizations were created with Matplotlib's Pyplot interface. Github was used as a tool for project management and version control, and the repository for this project can be found here: <https://github.com/ambrrrr/bioengc142finalproj>

To expand on the previous paper's study, I used the dataset for organic molecules on the GDB-11 database, which is preliminarily filtered by chemical stability, synthetic feasibility considerations, and simple valency rules¹. Further, molecules in the dataset were converted to 3D structures using the RDKit software package¹. This project focuses on organic molecules with four atom types (H, C, N, and O) and restricted the dataset on near equilibrium conformations.

Different ANN architectures were explored. One model attempt used GELU activation instead of ReLU, as the original paper stated the use of a Gaussian activation function¹. Another model attempt had 4 hidden layers with an order of 384, 160, 128, 96, 64, and 1, instead of 1 hidden layer like the original AtomicNet which had an order of 384, 128, and 1. Ultimately, the original AtomicNet model architecture was used, resulting in a total of 197636 parameters. Notably, this parameter count was more than the 124033 optimizable parameters per each individual atomic number neural network potential in the original paper¹.

For every mini batch of 1024 molecules, the network facilitated the following process: the input vector was fed forward to the output, the energy prediction was compared with the calculated ground truth DFT energy to compute Mean Squared Error (MSE) Loss with L2 Regularization, and backpropagation was performed. The Adam

Optimizer was used to update the weights and biases of the neurons of the ANN. Early stopping was implemented to prevent overfitting and improve efficiency in model training⁴. Manual hyperparameter selection was performed for L2 regularization, learning rate, batch size, and number of epochs. The following hyperparameter values were evaluated: L2 regularization values of 0.0, 1e-6, 1e-5, and 1e-4; learning rates of 1e-4, 5e-4, 1e-3, and 1e-2; batch sizes of 1024 and 8192; and epoch sizes of 10, 15, 20, and 50.

L2 Regularization	MAE	Learning Rate	MAE	Batch Size	MAE	Number of epochs	MAE
1e-4	2.72	1e-2	7.83			50	1.30
1e-5	1.63	1e-3	1.22	8192	0.80	20	0.77
1e-6	1.22	5e-4	0.80	1024	0.77	15	0.76
0.0	1.85	1e-4	2.27			10	0.70

Table 1-4 (left to right). Manual hyperparameter training. Table 1: Test MAEs for various L2 Regularization values; with original AtomNet architecture. Table 2: Test MAEs for various learning rate values; with original AtomNet architecture and L2 regularization hyperparameter value of 1e-6. Table 3: Test MAEs for various batch size values; with original AtomNet architecture, L2 regularization hyperparameter value of 1e-6, and learning rate parameter value of 5e-4. Table 4: Test MAEs for various epoch size values; with original AtomNet architecture, L2 regularization hyperparameter value of 1e-6, and learning rate parameter value of 5e-4, and batch size parameter value of 1024.

Multiple runs of models and N-fold cross-validation were also performed to assess model performance, stability, and signs of overfitting.

Results and Discussion

The model that provided the smoothest learning curves and smallest gap between training and validation curves was the original AtomNet architecture with 1 hidden layer and an order of 384, 128, and 1, with early stopping, L2 regularization of 0.0, learning rate of 1e-3, batch size of 8192, and 20 epochs (Figure 2). This model has better generalization and stability during training, which may be preferred since the stability and robustness of the model during training are critical⁵.

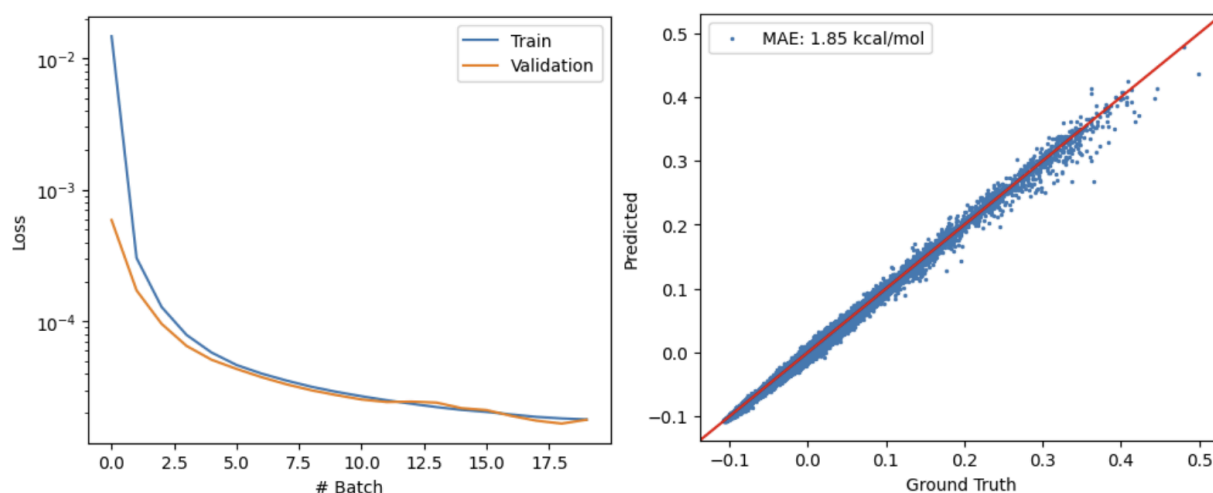


Figure 2: Learning curve and predicted energies vs the true energies for the model with the smoothest learning curves and smallest training/validation gap.

The model that provided the lowest MAE value was the original AtomNet architecture with 1 hidden layer and an order of 384, 128, and 1, with early stopping, L2 regularization of $1e-6$, learning rate of $5e-4$, batch size of 1024, and 10 epochs. This model has the highest predictive accuracy, making it more suitable for precise predictions of molecular energy values⁵. However, there is instability in the learning curves that could be caused by overfitting or ineffective hyperparameter values or architectural imperfections.

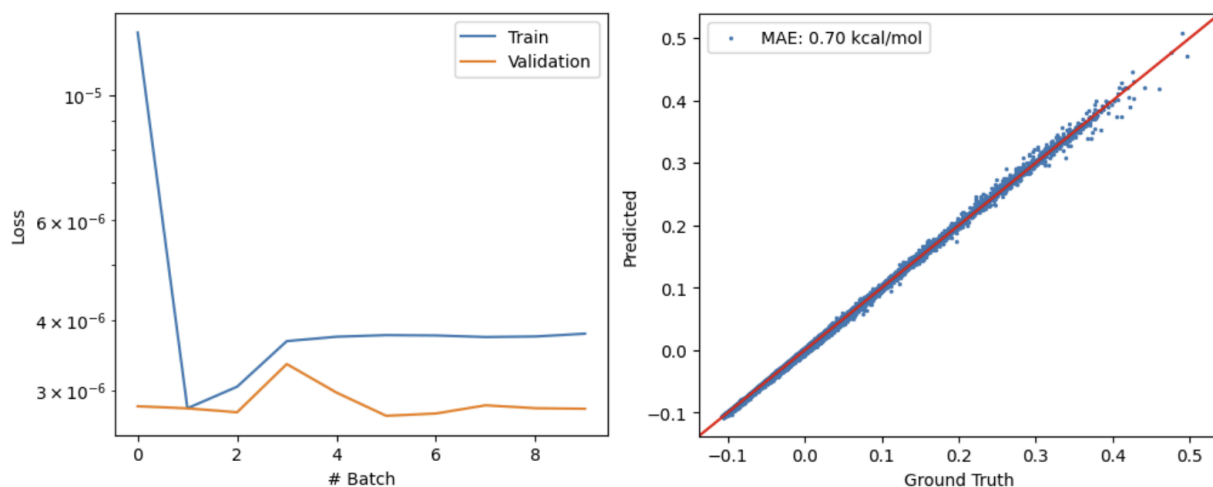


Figure 3: Learning curve and predicted energies vs the true energies for the model with the lowest MAE.

Ultimately, both models performed well and their RMSEs would be lower than the RMSE from the original paper, indicating that the artificial neural network (ANN) model accurately predicted the energies of molecules.

For future work, improvements to the training of the model can be made through further exploration of hyperparameter tuning and architectures. Specifically, time could be dedicated to addressing the variance and instability in evaluation. Additionally, this project could be enriched further by applying datasets with more heavy atoms, which would provide more generalization and applicability. In this project, all four heavy atoms (HA) were trained together, but evaluation of the heavy atoms separately could provide additional important insight.

References

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